

Computer Graphics

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Computer Science and Engineering
Metaverse Convergence
Graduate School

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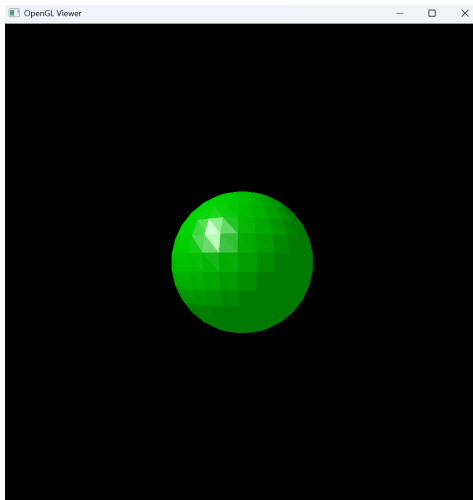
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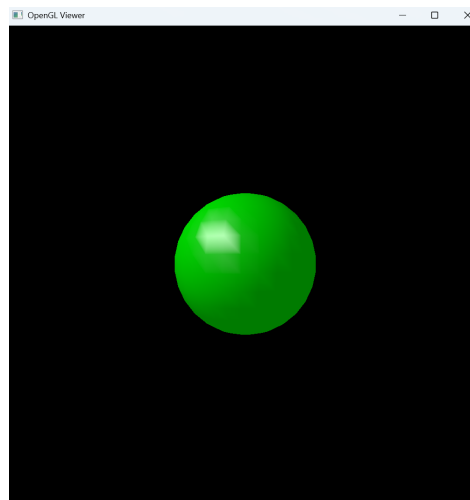
Assignment 6

Implementing a Rasterizer

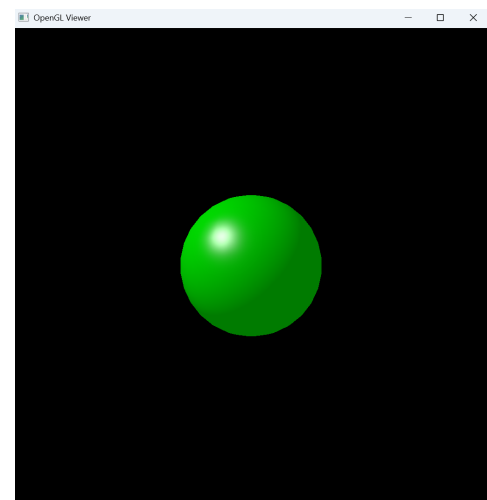
- Objective
 - The goal of this assignment is to write a simple rasterization pipeline and use it to draw a single shaded sphere, which is in turn represented using multiple triangles.
 - These triangles (and their vertices) are generated using code that is provided on the course webpage.
 - For reference, some views of the scene are shown below.



Flat Shading



Gouraud Shading



Phong Shading

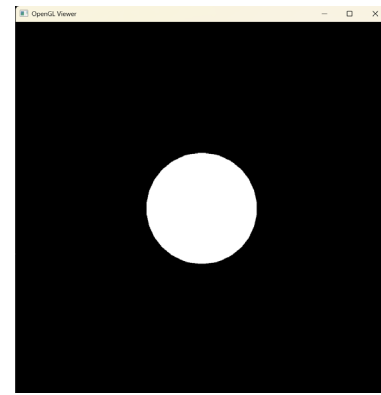
Questions

Q: Shading (3 Questions)

- Use the following sequence of transformations to display the sphere:
 - The provided code, “sphere_scene.cpp”, generates triangles for a unit sphere.
 - Use a modeling transform to transform the unit sphere to a sphere centered at $(0, 0, -7)$ with a radius of 2.
 - Set the eye point at $e = (0, 0, 0)$
 - Set the orientation to $u = (1, 0, 0)$, $v = (0, 1, 0)$ and $w = (0, 0, 1)$.
 - (Note that the camera is looking along the direction $-w$.)
 - Use a perspective projection transform with $l = -0.1$, $r = 0.1$, $b = -0.1$, $t = 0.1$, $n = -0.1$, $f = -1000$.
 - Use a viewport transform with $n_x = 1024$, and $n_y = 1024$.



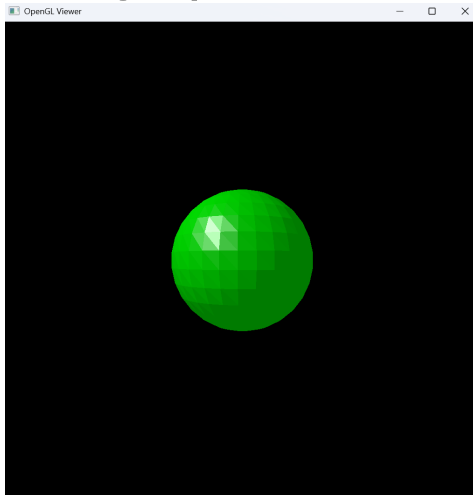
Q: Shading (3 Questions)



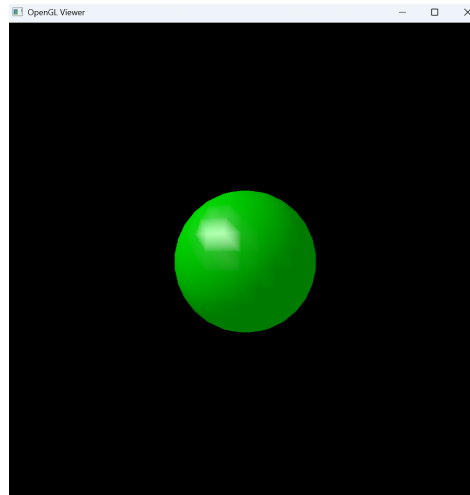
- Use the transformations and rasterization in HW5
 - Use the results from HW5
 - Update your HW5 results if they differ from the provided figure:
- Use the following material properties to shade the sphere:
 - Shade using Blinn-Phong reflection model:
 - $L = k_a I_a + k_d I \max(0, \mathbf{n} \cdot \mathbf{l}) + k_s I \max(0, \mathbf{n} \cdot \mathbf{h})^p$
 - $k_a = (0, 1, 0)$, $k_d = (0, 0.5, 0)$, $k_s = (0.8, 0.8, 0.8)$, $p = 64$.
 - Assume a single point light source at $(-4, 4, -3)$, emitting white light with unit intensity and no falloff.
 - Assume a white ambient light with an intensity of $I_a = 0.2$.
 - Apply gamma correction with $\gamma = 2.2$ in each case.

Q: Shading (3 Questions)

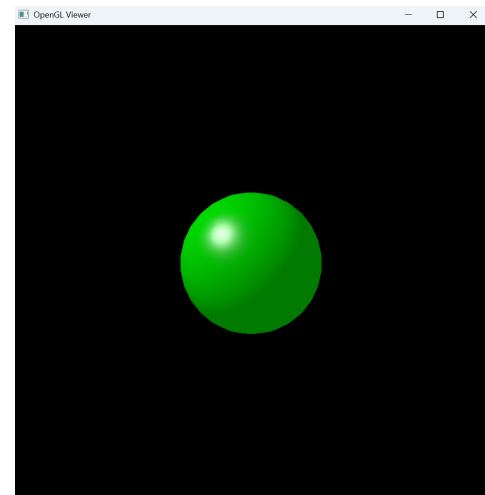
- Q1: Implement Flat Shading
 - (evaluate shading at the centroid of each triangle)
- Q2: Implement Gouraud (per-vertex) Shading
- Q3: Implement Phong (per-pixel) Shading
- Display your results as shown here.
 - Your output must be identical to these reference images.
 - Please be careful with floating-point calculations. Otherwise, you may get slightly different results!



Flat Shading



Gouraud Shading



Phong Shading



Q: Shading (3 Questions)

- In your presentation, you must clearly explain and demonstrate your implementation of the following:
 - Shading computation using per-triangle normals in Flat shading
 - Shading computation using per-vertex normals in Gouraud shading
 - Shading computation using per-fragment normals in Phong shading



Q4: Implement a New Feature

- Implement and demonstrate a custom feature
 - For example:
 - Using other shading models (Deferred shading)
 - Adding differently shaped objects (other than spheres)
 - Adding multiple objects or moving objects over time
 - Using different camera models, such as orthographic projection
 - This question will be graded on a relative scale (A/B/C).
 - Fewer than 50% of students will receive an 'A' grade.



Instructions

Submission Guidelines

- Important Notes

- Your code must include a separate project for each question but with a single solution.
- You are not allowed to use OpenGL to perform hardware rasterization.
 - You must implement your own transformation pipeline, rasterizer, and shading algorithms.
- Along with your code, please submit output screenshots for each question and explain them during your presentation.



Submission Guidelines

- Submission Requirements
 - Presentation video: Under 5 minutes and 30MB maximum (MP4 format)
 - Presentation slides: PDF format
 - Source code + output screenshots: VS 2022 project and image files (ZIP format)



Summary

- Assignment 6
- Questions
- Instructions



End of Class

Thank you

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